🏦 Banking Risk & Portfolio Analysis

End-to-End Data Analytics Project using Python, Exploratory Data Analysis & Power BI

Project Overview

Banking institutions manage large volumes of customer and financial data related to loans, deposits, account balances, income, and banking relationships. Analyzing this information helps financial institutions understand portfolio performance, identify concentration patterns, monitor financial exposure, and support informed decision-making.

This project analyzes a banking portfolio containing **2,530 clients**, with approximately **3.80B in total loans** and **3.25B in total deposits**. The complete analytics workflow includes **Exploratory Data Analysis (EDA) using Python in Google Colab, statistical and correlation analysis, customer and financial segmentation, and interactive dashboard development in Power BI**.

The objective was to transform banking data into meaningful financial and portfolio insights that help evaluate lending exposure, deposit strength, customer segments, and overall portfolio risk.

Business Problem

Banks need to maintain an appropriate balance between lending activity and available deposits while understanding which customer segments contribute the greatest financial exposure.Without effective analysis, it can be difficult to identify portfolio concentration, compare loan and deposit distributions, evaluate banking relationships, and detect potential funding pressure.

The key business challenge is to analyze the bank's portfolio and answer questions such as which customer segments contribute the highest loans and deposits, how financial exposure varies across different banking relationships, and whether the overall loan-to-deposit position indicates potential portfolio risk.This project addresses these challenges through **Python-based exploratory analysis and an interactive Power BI dashboard**, providing a consolidated view of the bank's financial health and portfolio composition.

Project Objectives

The primary objectives of this project are:

- Analyze the overall banking portfolio across **2,530 clients**.
- Examine the distribution of loans and deposits across customer segments.
- Analyze portfolio performance by **banking relationship, income band, nationality, and occupation**.
- Perform Exploratory Data Analysis using Python to understand financial distributions and relationships.
- Conduct statistical and correlation analysis between important financial variables.
- Evaluate the bank's **Loan-to-Deposit Ratio (LDR)** and loan-deposit funding gap.

- Identify segments contributing the highest loan exposure and strongest deposit base.
- Develop an interactive **multi-page Power BI dashboard** for portfolio monitoring.
- Generate meaningful financial and portfolio insights to support data-driven decision-making.

Dataset Overview

Attribute	Details
Dataset Name	**Banking Portfolio Dataset**
Business Domain	**Banking & Financial Analytics**
Total Records	**2,530 Clients**
Primary Identifier	**Client ID**
Total Loan Portfolio	**3.80B**
Total Deposit Portfolio	**3.25B**
Loan-to-Deposit Ratio (LDR)	**116.93%**
Loan-Deposit Gap	**549.57M**
Analysis Environment	**Google Colab**
Final Output	**Interactive Power BI Dashboard with Portfolio & Risk Insights**

Dataset Features

The dataset contains customer, banking relationship, and financial information including:

- **Client ID**
- **Name**

- Age
- Gender
- Nationality
- Occupation
- Income
- Income Band
- Banking Relationship
- Institution Advisor
- Joining Year
- Bank Loans
- Business Lending
- Credit Card Balance
- Bank Deposits
- Checking Account Balance
- Savings Account Balance
- Foreign Currency Account Balance

These attributes were used to analyze customer profiles, loan exposure, deposit distribution, banking relationships, financial segmentation, and overall portfolio performance.

Exploratory Data Analysis (EDA) Using Python

Exploratory Data Analysis was performed using Python in Google Colab to understand the structure, quality, distributions, and relationships within the banking dataset. Pandas and NumPy were used for data analysis and manipulation, while Matplotlib and Seaborn were used for data visualization.

Data Understanding and Inspection

The dataset was examined to understand its dimensions, column structure, data types, and overall composition. Initial inspection helped identify the available customer demographic, banking relationship, and financial variables required for further analysis.

Dataset Preview / `head()`

```
df = pd.read_csv("/content/clients_clean.csv")
df.head(5)
```

	Client ID	Name	Age	Location ID	Joined Bank	Banking Contact	Nationality	Occupation	Fee Structure	Loyalty Classification	...	Bank Deposits	Checking Accounts	Saving Accounts	Foreign Currency Account
0	IND81288	Raymond Mills	24	34324	06/05/19	Anthony Torres	American	Safety Technician IV	High	Jade	...	1485828.64	603617.88	607332.46	12249.96
1	IND65833	Julia Spencer	23	42205	10/12/01	Jonathan Hawkins	African	Software Consultant	High	Jade	...	641482.79	229521.37	344635.16	61162.31
2	IND47499	Stephen Murray	27	7314	25/01/10	Anthony Berry	European	Help Desk Operator	High	Gold	...	1033401.59	652674.69	203054.35	79071.78
3	IND72498	Virginia Garza	40	34594	28/03/19	Steve Diaz	American	Geologist II	Mid	Silver	...	1048157.49	1048157.49	234685.02	57513.65
4	IND60181	Melissa Sanders	46	41269	20/07/12	Shawn Long	American	Assistant Professor	Mid	Platinum	...	487782.53	446644.25	128351.45	30012.14

5 rows × 25 columns

Dataset Information / `info()`

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 25 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Client ID                 3000 non-null   object 
 1   Name                      3000 non-null   object 
 2   Age                       3000 non-null   int64  
 3   Location ID               3000 non-null   int64  
 4   Joined Bank               3000 non-null   object 
 5   Banking Contact           3000 non-null   object 
 6   Nationality               3000 non-null   object 
 7   Occupation                3000 non-null   object 
 8   Fee Structure             3000 non-null   object 
 9   Loyalty Classification    3000 non-null   object 
 10  Estimated Income          3000 non-null   float64
 11  Superannuation Savings    3000 non-null   float64
 12  Amount of Credit Cards    3000 non-null   int64  
 13  Credit Card Balance       3000 non-null   float64
 14  Bank Loans                3000 non-null   float64
 15  Bank Deposits             3000 non-null   float64
 16  Checking Accounts         3000 non-null   float64
 17  Saving Accounts           3000 non-null   float64
 18  Foreign Currency Account  3000 non-null   float64
 19  Business Lending          3000 non-null   float64
 20  Properties Owned          3000 non-null   int64  
 21  Risk Weighting            3000 non-null   int64  
 22  BRId                      3000 non-null   int64  
 23  GenderId                  3000 non-null   int64  
 24  IAId                      3000 non-null   int64  
dtypes: float64(9), int64(8), object(8)
memory usage: 586.1+ KB
```

Data Quality Assessment

Data quality checks were performed to identify missing values, duplicate records, inconsistent data types, and potential issues that could affect the reliability of the analysis.

The assessment ensured that the dataset was suitable for further exploratory analysis and visualization.

Missing Values and Duplicate Records Check


```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 25 columns):
 #   Column                    Non-Null Count  Dtype  
---  ------                    --------------  -----  
 0   Client ID                 3000 non-null   object 
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 2   Age                       3000 non-null   int64  
 3   Location ID               3000 non-null   int64  
 4   Joined Bank               3000 non-null   object 
 5   Banking Contact           3000 non-null   object 
 6   Nationality               3000 non-null   object 
 7   Occupation                3000 non-null   object 
 8   Fee Structure             3000 non-null   object 
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 17  Saving Accounts           3000 non-null   float64
 18  Foreign Currency Account  3000 non-null   float64
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 24  IAId                      3000 non-null   int64  
dtypes: float64(9), int64(8), object(8)
memory usage: 586.1+ KB
```

Descriptive Statistical Analysis

Descriptive statistics were generated for numerical variables to understand their overall distribution, central tendency, range, and variability.

The analysis provided an overview of important financial variables including income, loans, deposits, savings accounts, checking accounts, business lending, and credit balances.

Descriptive Statistics Output

```python
# Generate descriptive statistics for the dataframe
df.describe()
```

erannuation Savings	Amount of Credit Cards	Credit Card Balance	Bank Loans	Bank Deposits	Checking Accounts	Saving Accounts	Foreign Currency Account	Business Lending	Properties Owned	Risk Weighting	BRId	GenderId	IAId
3000.000000	3000.000000	3000.000000	3.000000e+03	3.000000e+03	3.000000e+03	3.000000e+03	3000.000000	3.000000e+03	3000.000000	3000.000000	3000.000000	3000.000000	3000.000000
25531.599673	1.463667	3176.206943	5.913862e+05	6.715602e+05	3.210929e+05	2.329084e+05	29883.529993	8.667598e+05	1.518667	2.249333	2.559333	1.504000	10.425333
16259.950770	0.676387	2497.094709	4.575570e+05	6.457169e+05	2.820796e+05	2.300078e+05	23109.924010	6.412303e+05	1.102145	1.131191	1.007713	0.500067	5.988242
1482.030000	1.000000	1.170000	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	45.000000	0.000000e+00	0.000000	1.000000	1.000000	1.000000	1.000000
12513.775000	1.000000	1236.630000	2.396281e+05	2.044004e+05	1.199475e+05	7.479440e+04	11916.542500	3.748251e+05	1.000000	1.000000	2.000000	1.000000	5.000000
22357.355000	1.000000	2560.805000	4.797934e+05	4.633165e+05	2.428157e+05	1.640866e+05	24341.190000	7.113147e+05	2.000000	2.000000	3.000000	2.000000	10.000000
35464.740000	2.000000	4522.632500	8.258130e+05	9.427546e+05	4.348749e+05	3.155750e+05	41966.392500	1.185110e+06	2.000000	3.000000	3.000000	2.000000	15.000000
75963.900000	3.000000	13991.990000	2.667557e+06	3.890598e+06	1.969923e+06	1.724118e+06	124704.870000	3.825962e+06	3.000000	5.000000	4.000000	2.000000	22.000000

Customer and Portfolio Segmentation

Customer characteristics were analyzed across demographic and banking-related dimensions to understand the composition of the portfolio.

The analysis considered variables such as:

- **Gender**
- **Nationality**
- **Occupation**
- **Banking Relationship**
- **Joining Year**
- **Income Band**

This segmentation helped identify how customers and financial activity were distributed across different groups.

Customer / Categorical Distribution Visualization

```
# Examine the distribution of unique categories in categorical columns
categorical_cols = df[["BRId", "GenderId","IAId", "Amount of Credit Cards", "Nationality", "Occupation", "Fee Structure", "Loyalty Classification", "Properties Owned"
for col in categorical_cols:
    print(f"Value counts{col}:")
    display(df[col].value_counts())
```

```
Value countsBRId:
```

BRId	count
3	1352
1	660
2	495
4	493

dtype: int64

```
Value countsGenderId:
```

GenderId	count
2	1512
1	1488

dtype: int64

```
Value countsIAId:
```

count

Financial Distribution Analysis

Major financial variables were analyzed to understand the distribution of customer balances and financial exposure across the portfolio.The analysis focused on variables related to:

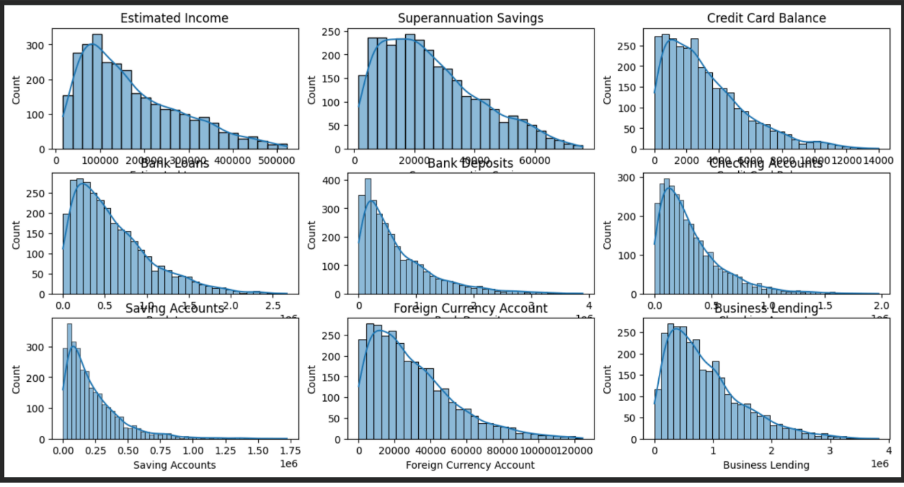
- Bank Loans
- Bank Deposits
- Business Lending
- Savings Accounts
- Checking Accounts
- Credit Card Balances
- Customer Income

The distributions helped identify patterns, variation, and concentration within the financial data.

Financial Variables Distribution Charts

```
numerical_cols = ['Estimated Income', 'Superannuation Savings', 'Credit Card Balance', 'Bank Loans', 'Bank Deposits', 'Checking Accounts', 'Saving Accounts', 'Foreign Currency Account', 'Business Lending']

#Univariate Analysis and Visualization
plt.figure(figsize=(15,10))
for i, col in enumerate(numerical_cols):
  plt.subplot(4,3,i+1)
  sns.histplot(df[col], kde = True)
  plt.title(col)
plt.show()
```



Income Band Analysis

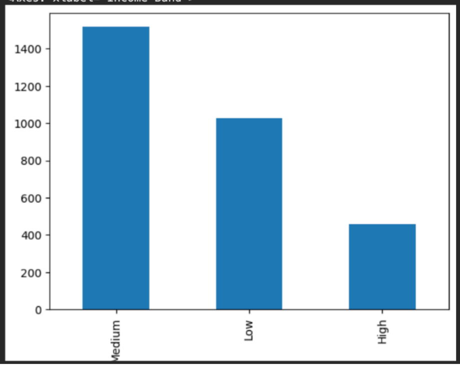
Customers were categorized into Low, Medium, and High Income Bands to enable meaningful comparison of financial activity across different income segments.Income segmentation was subsequently used to analyze the distribution of loans and deposits and understand the contribution of different income groups to the overall banking portfolio.

Income Band Distribution / Analysis

```
bins = [0, 100000, 300000, float('inf')]
labels = ['Low', 'Medium', 'High']
df['Income Band'] = pd.cut(df['Estimated Income'], bins=bins, labels=labels ,right = False)
```

```
df['Income Band'].value_counts().plot(kind = 'bar')
```

```
<Axes: xlabel='Income Band'>
```



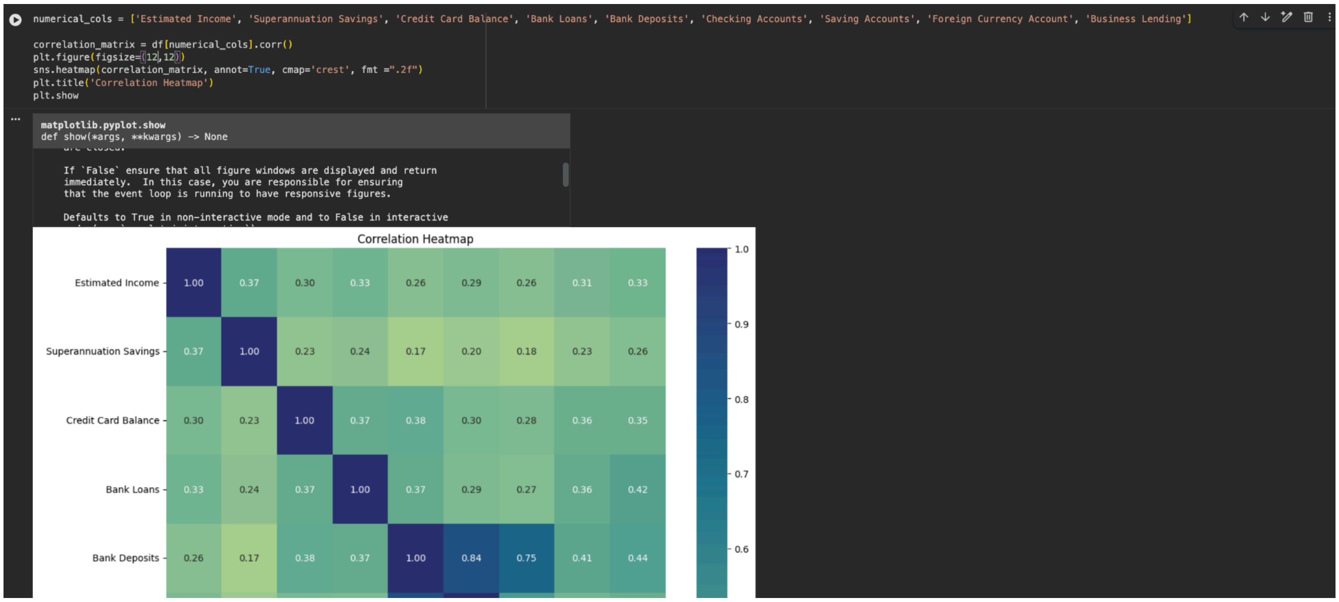
Correlation Analysis

Correlation analysis was performed to examine relationships between numerical financial variables and identify variables that move together within the banking portfolio.

A correlation heatmap was used to visually represent the strength and direction of relationships among variables such as loans, deposits, income, savings balances, checking balances, and other financial measures.

This analysis provided additional understanding of relationships within the financial data and supported the selection of relevant metrics for further portfolio analysis.

Correlation Heatmap



EDA Outcome

The exploratory analysis provided a structured understanding of the banking dataset and highlighted important patterns across customer characteristics, financial distributions, income segments, lending exposure, and deposit behavior.

The findings from the EDA formed the analytical foundation for developing the Power BI dashboard, where the portfolio was further analyzed through interactive KPIs, segmentation, loan and deposit analysis, and portfolio-level financial indicators.

Power BI Dashboard Development

After completing the Exploratory Data Analysis, the banking data was analyzed and visualized using Microsoft Power BI. An interactive four-page dashboard was developed to provide a comprehensive view of the bank's financial portfolio, loan exposure, deposit distribution, and portfolio-level risk indicators.

The dashboard consists of four analytical pages:

- **Financial Health Overview**
- **Loan Portfolio Analysis**
- **Deposit Portfolio Analysis**
- **Risk & Portfolio Summary**

Interactive filters and slicers were incorporated to enable analysis across dimensions such as banking relationship, joining year, gender, and institutional advisor.

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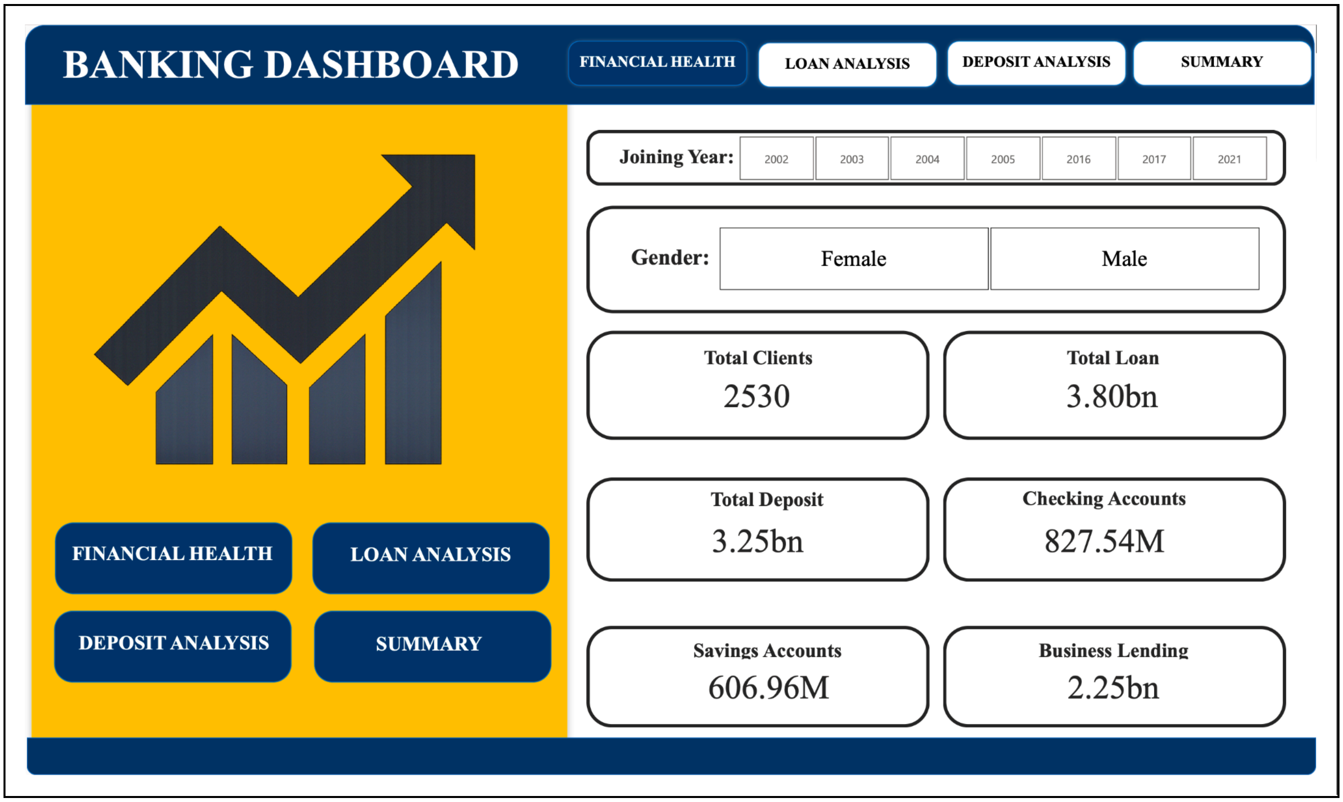
1. Financial Health Overview

The Financial Health Overview provides a high-level snapshot of the banking portfolio and its major financial indicators.

Key KPIs

- **Total Clients: 2,530**
- **Total Loans: 3.80B**
- **Total Deposits: 3.25B**
- **Checking Accounts: 827.54M**
- **Savings Accounts: 606.96M**
- **Business Lending: 2.25B**

This page serves as the dashboard's executive overview, allowing users to quickly understand the overall scale and financial composition of the banking portfolio.



2. Loan Portfolio Analysis

The Loan Portfolio Analysis page provides a detailed view of the bank's lending exposure across different customer and business segments.

Key KPIs

- **Total Loans: 3.80B**
- **Bank Loans: 1.54B**
- **Business Lending: 2.25B**
- **Credit Card Balance: 8.32M**

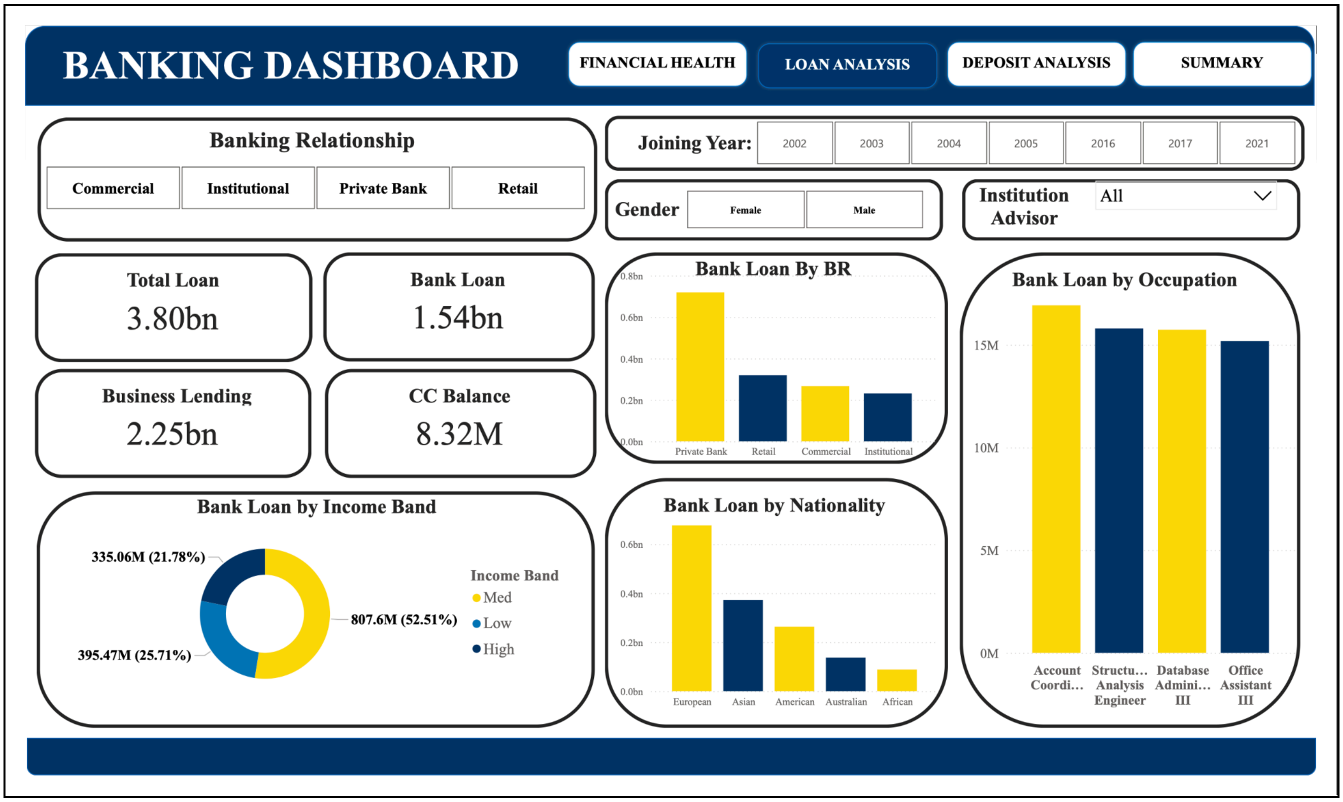
Loan exposure was analyzed across:

- **Banking Relationship**
- **Income Band**
- **Nationality**

- Occupation

The analysis enables identification of customer segments contributing significantly to the overall lending portfolio and helps highlight areas of portfolio concentration.

Interactive slicers allow users to further analyze lending patterns by banking relationship, joining year, gender, and institutional advisor.



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3. Deposit Portfolio Analysis

The Deposit Portfolio Analysis page evaluates the bank's deposit base and identifies the customer segments contributing most significantly to deposits.

Key KPIs

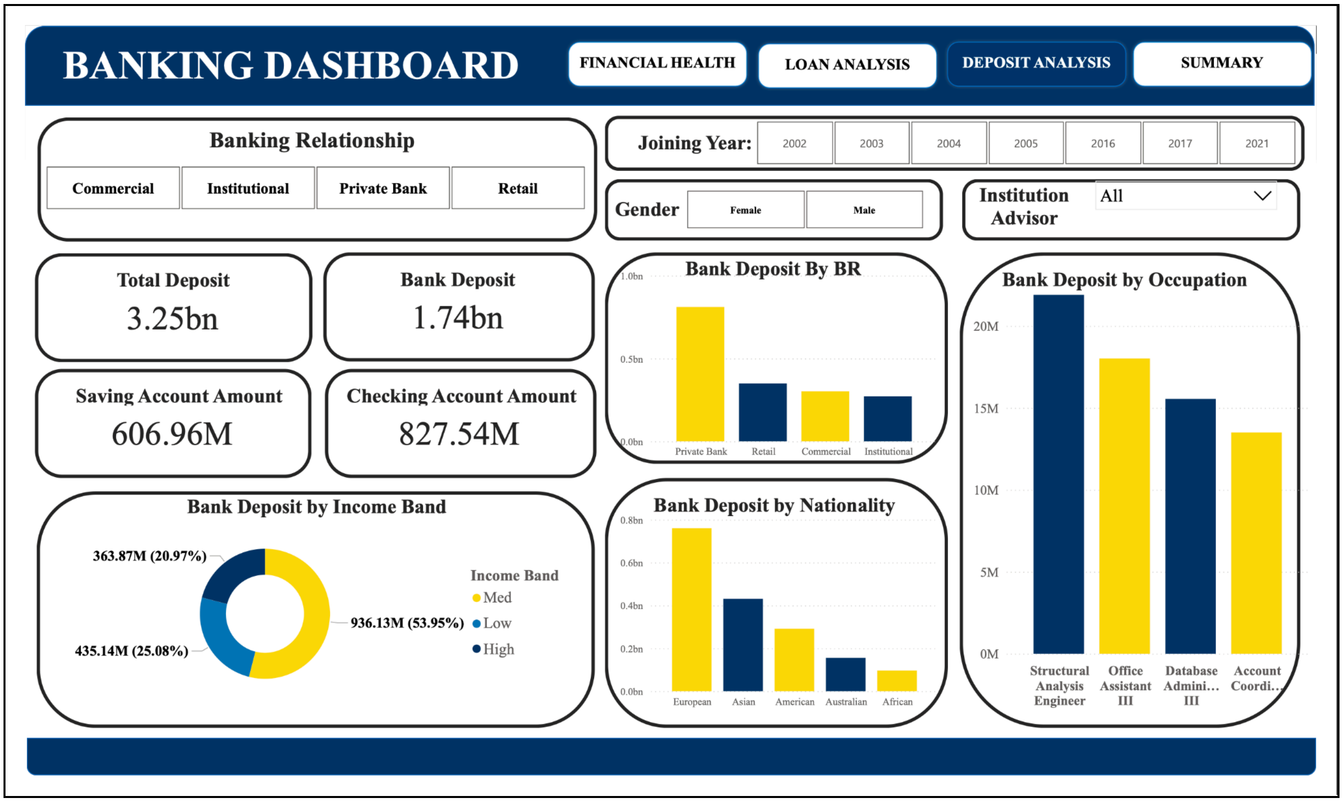
- **Total Deposits: 3.25B**
- **Bank Deposits: 1.74B**
- **Savings Account Amount: 606.96M**
- **Checking Account Amount: 827.54M**

Deposit distribution was analyzed across:

- **Banking Relationship**
- **Income Band**
- **Nationality**
- **Occupation**

This analysis provides visibility into the composition of the bank's deposit base and highlights segments that represent important sources of deposited funds.

Interactive filters enable detailed comparison across different customer and banking segments.



4. Risk & Portfolio Summary

The Risk & Portfolio Summary consolidates key lending and deposit indicators to provide an executive-level view of the portfolio's financial position.

Key Portfolio Indicators

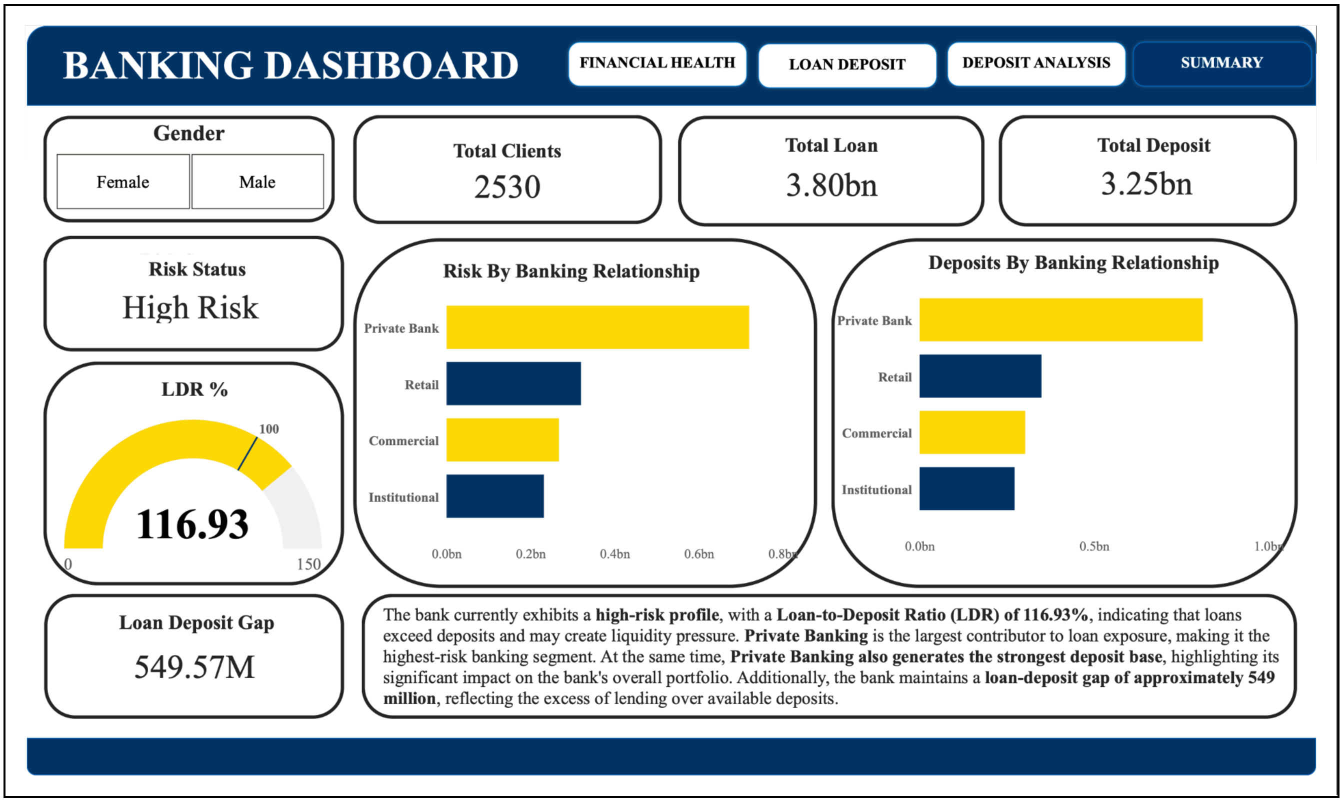
- **Total Clients: 2,530**

- Total Loans: 3.80B
- Total Deposits: 3.25B
- Loan-to-Deposit Ratio (LDR): 116.93%
- Loan-Deposit Gap: 549.57M
- Risk Status: High Risk

The Loan-to-Deposit Ratio of 116.93% indicates that the value of loans exceeds the deposit base represented in the portfolio. The portfolio also shows a loan-deposit gap of approximately 549.57M.

The page compares loans and deposits across banking relationships, highlighting Private Banking as the largest contributor to both loan exposure and the deposit base.

This summary provides a consolidated view for monitoring portfolio balance, concentration, and financial exposure.



Dashboard Outcome

The four-page Power BI dashboard transforms banking data into an interactive analytical solution that enables users to monitor financial KPIs, lending exposure, deposit composition, customer segmentation, and portfolio-level risk indicators.

The dashboard provides a centralized view of the banking portfolio and supports faster identification of financial patterns and areas requiring closer monitoring.

Key Business Insights

The analysis revealed several important findings about the banking portfolio:

1.) Total Loans Exceed Total Deposits

The portfolio contains approximately 3.80B in total loans compared with 3.25B in total deposits, resulting in a loan-deposit gap of approximately 549.57M.

This indicates that the value of lending exceeds the deposit base represented in the analyzed portfolio.

2.) Loan-to-Deposit Ratio Indicates Higher Portfolio Exposure

The calculated Loan-to-Deposit Ratio (LDR) is 116.93%.

An LDR above 100% indicates that total loans exceed total deposits in the analyzed dataset, highlighting the importance of monitoring the balance between lending activity and deposit funding.

3.) Private Banking Has the Highest Loan Exposure

Among the analyzed banking relationships, Private Banking represents the largest loan exposure.

This makes Private Banking an important segment for portfolio monitoring because a significant portion of lending activity is concentrated within this relationship category.

4.) Private Banking Also Provides the Strongest Deposit Base

Private Banking contributes the highest deposit amount among the analyzed banking relationships.

This indicates that the segment plays an important role on both sides of the portfolio, contributing significantly to both lending exposure and deposit funding.

5.) Medium-Income Customers Represent a Significant Portfolio Segment

Income-band analysis shows that medium-income customers contribute substantially to both loans and deposits.

This segment represents an important customer group for understanding portfolio composition and financial activity.

6.) Portfolio Exposure Varies Across Customer Segments

Analysis by nationality, occupation, income band, and banking relationship reveals differences in loan and deposit concentration across customer groups.

These dimensions provide management with greater visibility into where financial activity is concentrated within the portfolio.

7.) Interactive Segmentation Enables Deeper Portfolio Monitoring

The dashboard allows users to filter the analysis by banking relationship, joining year, gender, and institutional advisor.

This enables stakeholders to move beyond portfolio-level totals and investigate financial performance across specific customer and banking segments.

[INSERT IMAGE – Risk & Portfolio Summary Dashboard / Key Insights View]

Business Recommendations

Based on the analysis, the following recommendations are proposed:

- **Monitor the Loan-to-Deposit Ratio regularly to track changes in the balance between lending and the deposit base.**
- **Reduce or closely monitor the loan-deposit gap by strengthening deposit growth and reviewing lending concentration where appropriate.**
- **Closely monitor Private Banking exposure due to its significant contribution to both loans and deposits.**
- **Focus on customer segmentation to identify changes in financial exposure across income groups, occupations, nationalities, and banking relationships.**
- **Strengthen deposit acquisition strategies in segments where loan exposure significantly exceeds deposit contribution.**
- **Use interactive portfolio dashboards for periodic monitoring to enable faster identification of changes in key financial indicators.**
- **Extend future analysis with credit-quality indicators, such as repayment behavior, delinquency, default history, or credit scores, where such data is available, to enable deeper credit-risk assessment.**

Analytics Workflow

The project followed a structured analytics workflow, starting from raw banking data and progressing through exploratory analysis, visualization, and business insight generation.

Phase	Activity Performed	Tool Used	Output
1. Data Collection	Imported the banking customer and financial dataset	CSV	Raw Dataset
2. Data Understanding	Examined dataset structure, columns, data types, and descriptive statistics	Python (Pandas)	Dataset Understanding
3. Data Quality Assessment	Reviewed missing values, duplicates, data consistency, and numerical variables	Python (Pandas)	Analysis-Ready Data
4. Exploratory Data Analysis	Analyzed categorical and numerical variables, customer characteristics, and financial distributions	Python, Pandas, Matplotlib, Seaborn	EDA Insights
5. Customer Segmentation	Created and analyzed income bands and other customer segments	Python (Pandas)	Customer Segments
6. Correlation Analysis	Examined relationships among loans, deposits, income, and other financial variables	Python, Seaborn	Correlation Insights
7. Dashboard Development	Created KPIs, interactive visualizations, filters, and four analytical dashboard pages	Power BI, DAX	Interactive Dashboard
8. Portfolio & Risk Analysis	Evaluated LDR, loan-deposit gap, financial exposure, and segment concentration	Power BI	Portfolio Risk Insights

9. Business Recommendations	Translated analytical findings into actionable recommendations	Python + Power BI	Strategic Recommendations

Tools & Technologies

Tool / Technology	Purpose
Python	Exploratory Data Analysis, data inspection, segmentation, and analytical processing
Pandas	Data manipulation, dataset exploration, descriptive analysis, and customer segmentation
NumPy	Numerical computations and analytical operations
Matplotlib	Visualization of numerical distributions and financial variables
Seaborn	Statistical visualizations, categorical analysis, and correlation heatmap
Google Colab	Development environment for Python-based Exploratory Data Analysis
Power BI	Interactive dashboard development, portfolio analysis, and business visualization
DAX	Creation of calculated measures and financial KPIs
Git & GitHub	Project documentation, version control, and portfolio presentation

Project Outcome

This project successfully demonstrates an end-to-end banking and financial analytics workflow using Python-based Exploratory Data Analysis and interactive Power BI reporting.

Starting with customer and financial data covering 2,530 banking clients, the project explored customer characteristics, financial distributions, income segments, and relationships among key banking variables before transforming the analysis into a four-page interactive Power BI dashboard.

The final analysis evaluated approximately 3.80B in loans against 3.25B in deposits, identifying a 116.93% Loan-to-Deposit Ratio and a 549.57M loan-deposit gap. The analysis also highlighted Private Banking as a major segment in terms of both lending exposure and deposit contribution.

The project demonstrates how data analytics can help financial institutions:

- **Monitor loan and deposit portfolios.**
- **Identify concentration across customer and banking segments.**
- **Evaluate portfolio-level financial indicators.**
- **Improve management reporting through interactive dashboards.**
- **Transform complex banking data into meaningful business insights.**
- **Support faster and more informed data-driven decision-making.**

Overall, the project demonstrates practical skills in Python, Exploratory Data Analysis, financial analytics, Power BI, DAX, data visualization, portfolio analysis, and business insight generation.